Student Loans in the United States

Jeffrey T. Denning, University of Notre Dame and NBER

Introduction

College causally increases wages but not everyone who wants to make this productive investment has access to savings or credit to pay for college (Hoekstra 2009; Zimmerman 2014; Mountjoy 2024 but Friedman 1955). Government is often involved in lending for education because human capital (unlike a house or a car) cannot be used as collateral, and thus credit will be underprovided by the private market. Loan repayments can impose financial burdens on students and families, with especially bad consequences when borrowers default (including wage garnishment and damaged credit).

In 2024, there is $1.59 billion of outstanding student debt up from under $400 million in 2005. Not only is the amount of debt large, it is widespread with approximately 38 million people having student debt.[1] Accompanying this increase in borrowing, there has been an increase in delinquency on debt. This rapid increase in the use of student loans has prompted a discussion of if student loans are accomplishing their intended goals—do they help students complete education and gain access to high paying jobs?

Key Findings

- *Increases in access to borrowing often improve student educational, labor market, and financial outcomes.* Borrowing may allow students to focus on schooling, avoid dropout, and improve academic outcomes. Borrowing increases arising from policy changes or nudging students to change borrowing improves student outcomes including improved grades, higher credits earned, transfer to four-year schools, increased probability of graduation, and even higher earnings. Increased borrowing reduces reliance on credit cards and reduces working for wages in college which likely helps explain the positive effects on student outcomes. Importantly, these results apply when borrowing increases occur while holding the cost of college fixed.
- *Increases in student debt arising from increasing costs of college reduce educational attainment and damage household finances.* When the price of college increases, students may use additional student loans to cover the increase. When this happens, it is best to think of this as the effects of an increase in cost of college, rather than the effects of increased

[1] Quarterly Report on Household Debt and Credit, 2024: Q2. https://www.newyorkfed.org/microeconomics/hhdc. https://www.newyorkfed.org/studentloandebt/. https://studentaid.gov/data-center/student/portfolio

student debt. Increases in student debt arising from increases in prices reduce educational attainment and negatively affects household finances.

- *Increasing access to student loans increases the price that students pay for college.* This effect is termed the "Bennett Hypothesis" which states that when loans become more available, schools increase their prices. This could happen because loans increase a students' willingness to pay for higher education. Recent studies have found evidence in support of this hypothesis; increasing access to student loans increases prices. This is consistent with other studies that show increases in other financial aid also increases prices.
- *Income Driven Repayment improves borrower financial health.* Income driven repayment changes a borrower's monthly payment to a fraction of their income, rather than paying a fixed amount. This is designed to act as insurance against income shocks that borrowers experience. A series of papers have shown that enrollment in income driven repayment plans, relative to the default fixed payment plan, reduces delinquency on debt, consistent with the intention of these policies. Additionally, there is some evidence that borrowers see improvements in their financial outcomes in the longer run.
- *The presentation of information about loan options has large impacts on loan use.* The way that borrowing options are presented affects the amount that students borrow. The default repayment plan also affects repayment plan choice. The most consistent finding is that the default option has outsize influence on student loan borrowing and repayment decisions.

Policy Background

The United States federal government has several loan programs. Undergraduate borrowers can borrow in two primary programs, Direct Subsidized Loans and Direct Unsubsidized Loans. Financial need does not determine the total amount a student can borrow; rather, this depends on a student's status as dependent (traditional student) versus independent (nontraditional student) and their year in school. Subsidized loans are available to students who demonstrate financial need and they currently have the same interest rate as unsubsidized loans. Subsidized loans do not accrue interest while the student is enrolled in college or for six months after enrollment ends. Federal loans have yearly and lifetime limits for undergraduates. Dependent first-year students can borrow up to $5,500 with no more than $3,500 being subsidized. Dependent second-year students can borrow up to $6,500 with no more than $4,500 being subsidized. Dependent third-year students and beyond can borrow up to $7,500 with no more than $5,500 being subsidized. In total, depend undergraduates can only borrow $31,000 in direct loans in their lifetime. Independent undergraduates have higher yearly and lifetime borrowing amounts.

Graduate students can borrow using direct loans (with a yearly limit of $20,500 and a lifetime limit of $138,500 with higher limits for some health programs). Additionally, graduate students can borrow up to the Cost of Attendance of their program using the Grad PLUS loan program with no lifetime maximum on borrowing.

Lastly, parents can borrow on behalf of their children using Parent PLUS loans. Parents can borrow if they do not have an adverse credit history and can borrow up to a student's Cost of Attendance less any other financial assistance they receive. Parents borrowing for their children present a different motivation for borrowing. Parents are not investing in their own human capital with an expectation that increases in their earnings can pay back any debt accumulated; rather, they may be more motivated by transferring income to future generations.

Eighty-five percent of borrowing is from federal loans (Ma and Pender 2023). Delinquency and default vary by student characteristics such as school with less selective schools generally having higher default rates (Looney and Yannelis 2015). Borrowers with small balances are more likely to default (Looney and Yannelis 2015). Borrowers with very high balances do not to default at as high of rates and accumulated much of their debt in graduate school (Looney and Yannelis 2018). Notably, default rates are higher for Black borrowers than for White borrowers (Scott-Clayton 2017).

Key Findings

Increases in access to borrowing often improve student educational, labor market, and financial outcomes.

Studying the effect of access to student loans on student enrollment, graduation, and earnings has been difficult because nearly all students are eligible for federal student loans. Hence, it is hard to find a good comparison group. Currently there is very little evidence on how much the availability of student loans affects the decision to *attend* college. We have more evidence regarding how loan availability affects students after they've made a decision to start college. Access to additional loans may help students complete college by allowing them to focus on their studies and or reduce working for wages during college.

Black et al. (2023) study the effect of increases in the annual maximum amount students could borrow that occurred in 2005-06 and 2007-08. They compare students who borrowed the maximum amount in their first year to students who borrowed less than the maximum amount before and after the change. This strategy inherently only reveals the effect of higher borrowing limits on students who were already enrolled in college. They subset their analysis into four groups defined by dependent versus independent students and students who initially entered a two-year versus four-year school. They find that dependent students who entered a initially entered a four-year school borrowed more, were more likely to graduate from college and had higher post college earnings. Increased borrowing limits reduced use of credit cards while in

college and earnings while in college. This suggests that student loans may allow students to focus on schooling and see improved outcomes. Dependents entering a two-year school also increased borrowing and saw some evidence of improved educational outcomes and earnings but the results are less precisely estimated. In contrast, independent students who have higher annual maxima did not increase their borrowing. The authors also find decreases in delinquency on debt using a national sample of borrowers and no effects of increased access to loans on mortgages.

Other work has considered four-year students. Denning (2019) studies an increase in financial aid for students at four-year schools using the age cutoff for independent students and a regression discontinuity design. For the poorest students in the sample (who previously had an Expected Family Contribution of 0) there was no increase in grant aid, but an increase in borrowing. This increase in borrowing led students to graduate earlier than they otherwise would. In Denning and Jones (2021) the authors use credits completed and a regression discontinuity design to compare students who are eligible for additional loans (e.g. freshman versus sophomores) and find no effect of increased access to student loans on grade point average, credits, persistence or graduation. The change in borrowing is much smaller in Denning and Jones (2021) than in Black et al. (2023) which make explain some of the discrepancy. Sun and Yannelis (2016) study banking deregulation and find that it caused an increase in borrowing and enrollment. This is notable because banking deregulation affected borrowing from private sources).

In the two-year sector there has been more work on the effects of student loans. This is because some schools opt out of offering student loans and others vary the way student loans offers are presented to students. Schools often make this choice for fear of becoming ineligible to disburse Title IV aid which includes Pell Grants and federal student loans. In recent years, very few institutions are subject to sanctions that would affect the ability to offer Title IV aid, but many schools either completely opt out of the student loan program or list an amount less than the maximum amount a student is eligible to borrow on their financial aid letter.

Wiederspan (2016) uses a two-way fixed effect strategy to identify the effect of a two-year school opting out of offering federal student loans. The author finds opting out decreased borrowing of loans and that students who borrowed attempted more credits and attempt and complete math and science courses. Similarly, Dunlop (2013) compares students at community colleges that offer and do not offer student loans and finds that high need and female students increase total months of enrollment and that dependent students increase bachelor's degree attainment.

Two other papers study community colleges and induced changes in borrowing via an intervention that changed the way the school presented student loans. Marx and Turner (2019)

randomly altered financial aid award letters so that some students saw the full amount they were eligible for and others saw $0. This is a common practice called "packaging" and did not change the amount students could borrow. This change induced differences in borrowing and, as a result, allowed the researchers to estimate the effect of additional student loans on attainment. The additional borrowing increased credits attempted, credits earned, GPA, and transfer to four-year colleges. Barr, Bird, and Castleman (2021) partnered with the Community College of Baltimore County and randomly sent messages to students prompting them to make active choices about their borrowing decision. This led students to borrow less, have worse academic outcomes, and default on students at higher rates. Interestingly, Barr, Bird, and Castleman (2021) and Marx and Turner (2019) use different "nudges" to change borrowing but come to similar conclusions about the benefits of increased borrowing.

Overall, the effects of offering students access to loans has been to generally increase attainment with one study finding reductions in delinquencies and increases in earnings. The evidence for the effects of loan access on attainment are strongest in the community college sector. The positive effects of loans are not uniform with (Denning and Jones 2021) finding no effects and (Black et al. 2023) finding no effects for independent students. Some evidence suggests that in the absence of access to student loans, students may turn to other ways to finance their education that reduce academic success. There is some evidence for this with increased availability of student loans reducing the use of credit cards and working during college. Further, some of the effect likely comes through increased persistence early on in a student's academic career (or via increased transfer for two-year students).

Increases in student debt arising from increasing costs of college reduce educational attainment and damage household finances

Increases in student debt could arise for several reasons. Consider two scenarios. First, tuition increases cause some students to borrow more. Second, the price of college is unchanged, but the maximum amount students can borrow increases causing some students to borrow more. In both of these cases, student debt has increased. However, in the first case, students are paying more for college and in the second case they are not. This distinction is important for thinking about policy. The prior section discusses the former case—what happens when students borrow more, holding price fixed. I will now discuss what happens when prices change (via grants or tuition) and how this affects students.

Chakrabarti et al. (2023) use credit data merged to enrollment and graduation data to study the effects of increases in tuition. They use shocks to tuition that affected some cohorts of students and not others. They find that increases in tuition increases borrowing. They then show that increases in tuition reduces graduation, mortgages balances, and increases credit card delinquency. Mezza et al. (2020) use similar data and instrument for tuition experienced with

tuition of the student's home state. They find that increases in tuition result in increases in student loans. They also find a reduction in the homeownership rate. These studies ask a different question than the effects of access to student loans but still lend insight into how students use student debt to insulate themselves against price shocks. This places them in a similar category to several other studies on changes in prices which generally find that reducing the price of college improves student outcomes. (Bettinger et al. 2019; Marx and Turner 2018; Denning, Marx, and Turner 2019; Scott-Clayton and Zafar 2019).

The findings from studies that consider increases in the price of college that increase student debt are notably quite different than studies that consider the effect of access to additional student loans. When students have to pay more for college, their educational and household finance outcomes generally get worse. When students are given access to additional loans holding price fixed, their educational, household finance, and earnings outcome generally improve.

Increasing access to student loans increases the price that students pay for college.

Increasing the supply of student loans may increase the price of college which has been termed the "Bennett Hypothesis" after former Secretary of Education William Bennett. There is increasing evidence for this. Lucca, Nadauld, and Shen (2019) study an increase in loan limits for undergraduates and find that the pass through of increases in tuition was 60 percent for subsidized loans and for 20 percent for unsubsidized loans with largest effects for costly degrees and at two year colleges and for profit institutions. Black, Turner, ⓡ Denning 2023 study the creation of Grad PLUS which increased the amount students could borrow for graduate school. Using variation in which programs were more exposed and they find increases in net prices of about 60 percent. Other research that studies the beginning of Grad PLUS that uses undergraduates as a comparison group does not find evidence of the Bennett Hypothesis (Kelchen 2019; 2020).

While the evidence for the Bennett Hypothesis is somewhat limited when considering loans alone, there is additional evidence that prices increase when aid increases. This includes Lucca, Nadauld, and Shen (2019) who also consider increases in Pell Grant aid as well as Turner (2017) who also find increases in prices paid by students as a result of the Pell Grant. Other examples include the Post 9/11 GI Bill (Baird et al. 2022) and eligibility for all Title IV aid (Cellini and Goldin 2014).

Income Driven Repayment improves borrower financial health

Students can repay federal student loans using a variety of plans. The default is a traditional plan that requires monthly payments of a fixed size over a time frame (the default is 10 years). Other repayment plans require a fraction of income (above a minimum income threshold).

These differences in repayment plans offer different amounts of insurance against bad income shocks. The fraction of borrowers using income based repayment plans has grown for undergraduates from 11 to 24 percent from 2010 to 2017 (CBO 2020).

Herbst (2023) uses data from a loan servicer and the insight that some call center agents are more effective at inducing borrowers to enroll in Income Driven Repayment (IDR). He finds that use of income driven repayment reduces delinquencies and suggestive evidence of increases in credit scores and mortgage holding rates. Mueller and Yannelis (2022) study a setting where Navient randomly sent prepopulated income driven repayment applications to borrowers who had applied to IDR. This treatment meaningfully increased enrollment in IDR. They then show that IDR enrollment reduced monthly payments, new delinquency on debt, and increased credit card balances.

Monarrez and Turner (2024) use Department of Education data and a policy discontinuity that means that very similar borrowers either have a zero payment or a positive payment amount. They show that qualifying for a minimum payment of zero for income driven repayment plans reduces delinquency and default in the short run but in the longer run this increases default. This suggests that administrative issues are important in determining repayment.

Repayment rates have changed substantially in recent years. Darolia, Monarrez, and Turner (2024) show that the fraction students repay five years after entering repayment dropped dramatically from 2004 to the 2014 cohorts. This suggests that future policy attention and research will likely focus on repayment.

Presentation of Information

Choosing how much to borrow is a complex decision that requires forecasting expected costs, assessing future income, etc. Given this complexity there are several studies that try to understand how borrowers make this choice. Marx and Turner (2019) show that packaging loans—the practice of including loans in financial aid offer letters—increases student loan take up. Kramer, Lamb, and Page (2021) randomly prepopulated acceptance forms about financial aid to either decline, accept, or make an active choice. Prepopulating a decline reduced borrowing by around 5 percent. Field (2009) shows that at NYU law school framing financial aid as either a loan or a reduction in tuition had meaningful impacts on take up. In one framing, the financial aid was a loan that would be forgiven if a student worked in public interest law versus as a reduction in tuition that would be have to be repaid if the student did not work in public interest law. Though financially equivalent, the tuition reduction meaningfully increased public interest law job placement and affected enrollment in the program. Marx and Turner (2020) sent emails to students at a community college informing them that they did not have to borrow the maximum amount which had no effect on borrowing decisions. Emails that

referenced amounts borrowed by recent graduates reduced borrowing. The authors conclude that students making choices about borrowing may face cognitive overload. Barr, Bird, and Castleman (2021) sent text messages to community college students to encourage active borrowing decisions and this led to reduced borrowing.

Similarly, choosing a repayment plan requires a complex calculation including forecasting future income, assessing risk, etc. Abraham et al. (2020) use a lab experiment to show that choice of repayment plans does not follow standard models of economic decision making. Rather, more behavioral models of choice describe the choices made by participants. Cox, Kreisman, and Dynarski (2020) use an experiment to study the choice of IDR plans and find that default options are quite important whereas information on the distribution of earnings has no effect.

The primary conclusion from these studies is the choice architecture of borrowing and repayment decision has meaningful influence on borrowing and repayment decisions. The most consistent finding is that the default option has outsize influence on the choices ultimately made by borrowers.

Recent Policy Issues Debt Forgiveness/Pause

One of the effects of student loans is the way in which repayment can affect outcomes. Recently, there have been calls for forgiveness of student loans and attempts by the Biden administration for broad forgiveness. Di Maggio, Kalda, and Yao (2019) study a situation where debt was discharged because a private student loan lender lost lawsuits that required student debt to be forgiven because it was not able to establish a chain of title. This debt forgiveness reduced delinquency on other accounts and increased geographic mobility and income. Dinerstein, Yannelis, and Chen (2024) study how a pause in payments for many borrowers affected credit outcomes by using the fact that the pause only applied to loans owned by the federal government. They find that pausing payments increased borrowing on other types of loans rather than reduce delinquencies.

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